Diagram 1

Figure 1: Diagram 1

FundFlow — Accounting & Ledger Reference

One page · For accountants and auditors

Chart of accounts (conceptual)

Account	Type	Invariant
Master Cash	Pool cash	Balance = Σ Member Cash
Master Fund	Pool equity	Balance = Σ Member Fund
Master Bank	Bank control	Tied to imported `bank_transactions`
Member Cash	Liability to member	Spendable balance
Member Fund	Member equity in pool	May be negative (active loan)
Loan sub-account	Principal tracking	Per active loan

Posting conventions

Mirror pairs (same transaction)

Event	Order	Legs
Cash inflow	Master first	CR master cash → CR member cash
Cash outflow	Member first	DR member cash → DR master cash
Fund credit	Master first	CR master fund → CR member fund
Fund debit	Member first	DR member fund → DR master fund

Do not attach `member_id` to master cash legs unless documented — it triggers auto-collection.

Bank clearance (non-posting)

Ledger records **intent**; clearance updates `is_cleared` and matches lines. **No additional cash/fund entries** on successful match.

Journal patterns by event

Contribution collection

	Account	Dr/Cr
Dr	Member Cash	amount
Dr	Master Cash	amount (mirror)
Cr	Member Fund	amount
Cr	Master Fund	amount (mirror)

Loan disbursement

	Account	Dr/Cr
Dr	Loan account	principal
Dr	Member Fund	member + master portions (mirrored on master fund)
Cr	Loan account	member portion applied to principal
Cr	Member Cash	full disbursement
Cr	Master Cash	mirror

Cash-out approval

	Account	Dr/Cr
Dr	Member Cash	amount
Dr	Master Cash	mirror
—	Uncleared bank transaction	negative placeholder

Loan repayment (with cash flow)

	Account	Dr/Cr
Cr	Member Cash	amount (cash-in mirror)
Cr	Master Cash	mirror
Dr	Member Cash	amount (collection)
Dr	Master Cash	mirror
Cr	Member Fund	amount
Cr	Master Fund	mirror
Cr	Loan sub-account	principal

Net effect on cash: **zero** (paired mirror); fund and loan reflect repayment.

Repayment economics

`Repayment obligation = master_portion + (amount_approved × settlement_threshold)`

- Member's own fund portion at disbursement = **equity**, not collected via EMI.
- **100% member-funded** loan (`master_portion = 0`, `settlement_threshold = 0`): no EMI schedule; completed at import.

Reconciliation checks

Code	Formula / rule
`MASTER_CASH_POOL_DRIFT`	`master_cash.balance` `Σ member_cash.balance`
`MASTER_FUND_POOL_DRIFT`	`master_fund.balance` `Σ member_fund.balance`
`MEMBER_CASH_DRIFT`	Component sum vs `accounts.balance` (§5.13)
`MEMBER_FUND_DRIFT`	Component sum vs `accounts.balance` (§5.13)

Repair: `php artisan accounting:rebuild-balances --tenants=<tenant>` when stored balance transaction net.

Legacy migration notes

Pattern	Risk
Contribution repair + raw `DELETE` on transactions	Cash balance drift — use `deleteTransaction()`
Bulk repayment import	Credit/debit pairs; rebuild balances after import
Implicit member-funded loans	Complete when no pool/settlement obligation

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Audit trail

Layer	Source
Ledger	`transactions` (type, amount, `balance_after`, reference)
Bank	`bank_transactions` + import `bank_statements`
Workflow	Contributions, loans, cash-outs, fund postings
Exceptions	`reconciliation_exceptions` (nightly batch)

Slide outline (presentation)

1. **Title** — FundFlow ledger architecture
2. **Accounts** — Master vs member; pool invariants
3. **Mirrors** — Posting order and pairing rules
4. **Journals** — Contribution, loan, cash-out, repayment
5. **Bank** — Intent vs clearance
6. **Repayment math** — Master slice + settlement
7. **Reconciliation** — Drift codes and repair
8. **Migration** — Known integrity patterns

Full diagrams and sequences: fund-flow-dynamics.md